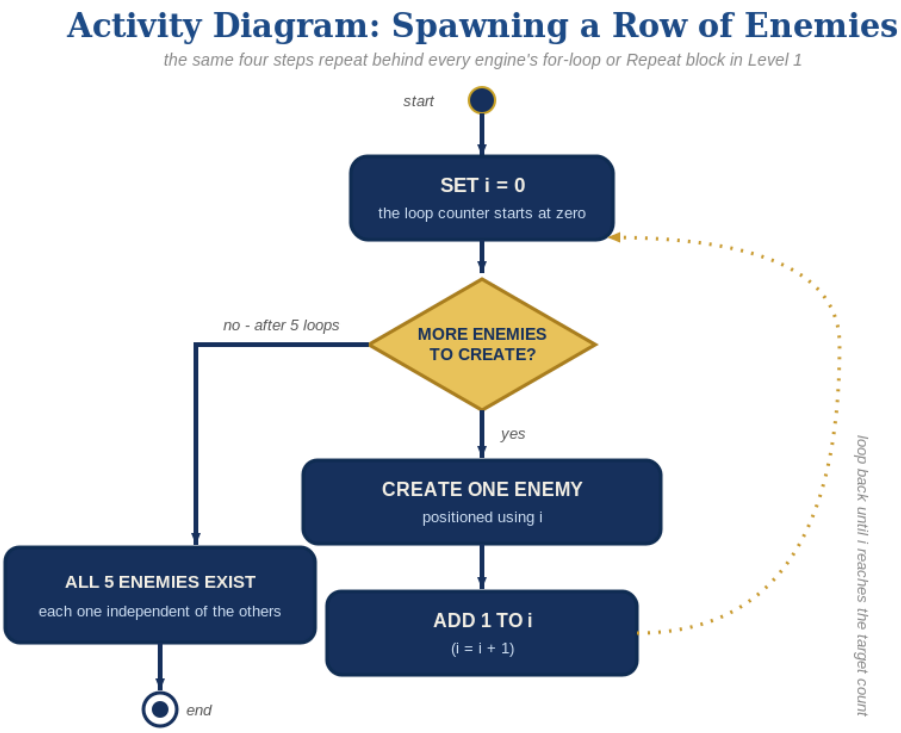
Lesson 5 Teacher's Key

Godot Engine

INSTRUCTOR ONLY - do not distribute to students

Level 1 - Complete Worked Solution



Every engine's Level 1 solution runs this same shape - a counter, a repeat check, one creation step, and an increment - even though the exact blocks or code syntax differ. Tracing a spawn bug means finding which of these four steps broke.

The spawn loop, in general - the same counter, repeat check, creation step, and increment behind Level 1 in every engine, whatever the exact blocks or syntax look like.

A for loop in the main scene's own script - enemy.tscn itself needs zero changes from Lesson 4:

```
# Main scene's script:
var enemy_scene := preload("res://enemy.tscn")

func _ready():
    for enemy_count in range(5):
        var new_enemy := enemy_scene.instantiate()
```

```
        new_enemy.position = Vector2(150 + enemy_count * 200, 50)
        add_child(new_enemy)
```

Level 2 - Complete Worked Solution

enemy.tscn's own script (its Hitbox's area_entered connection and hit-response) is fully self-contained, so every instance the loop creates behaves independently and correctly with no extra code needed for Level 2.

Level 3 - All Worked Solutions

Every stretch-goal idea from the handout, fully worked - not just one. Useful for validating each idea actually works before class, and as a ready reference no matter which one a student ends up picking.

Sound: already works

Every spawned enemy carries its own Hit AudioStreamPlayer2D from Lesson 4 - no changes needed.

Visual: striped row

Set modulate based on the loop index right after instancing, from the spawner's own script. Color(0.8, 0.1, 0.2) is a deep crimson red (mostly red, small amounts of green/blue) and Color(0.95, 0.4, 0.5) is a lighter dusty rose/pink (same red-dominant mix, just brighter with more green/blue mixed in) - alternating them produces a row where every other enemy is a visibly different shade of red, not a blended new color:

```
for enemy_count in range(5):
    var new_enemy := enemy_scene.instantiate()
    new_enemy.position = Vector2(150 + enemy_count * 200, 50)
    if enemy_count % 2 == 0:
        new_enemy.modulate = Color(0.8, 0.1, 0.2)
    else:
        new_enemy.modulate = Color(0.95, 0.4, 0.5)
    add_child(new_enemy)
```

Structure: spawner vs. behavior stay separate

The main scene's _ready() only ever creates and positions enemies - it never reaches into an enemy's hit-response logic, which stays entirely inside enemy.tscn's own script.

Complete Level 3 solution (all 3 goals combined)

The spawner loop with the visual idea added - sound needs no changes since it already lives on each enemy instance:

```
var enemy_scene := preload("res://enemy.tscn")

func _ready():
    for enemy_count in range(5):
        var new_enemy := enemy_scene.instantiate()
        new_enemy.position = Vector2(150 + enemy_count * 200, 50)
        if enemy_count % 2 == 0:
            new_enemy.modulate = Color(0.8, 0.1, 0.2)
        else:
            new_enemy.modulate = Color(0.95, 0.4, 0.5)
        add_child(new_enemy)
```